

ADITYA PATEL

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EDUCATION

Bachelor of Science in Computing Science

University of Alberta

Edmonton, AB

Expected Graduation: 2027

Relevant Courses: Algorithms | Machine Learning | Software Engineering | Database Management | Computer Organization & Architecture

EXPERIENCE

Full Stack Developer Intern

Jarillo Consulting

Edmonton, AB

Jan 2026 - April 2026

- Accelerated feature delivery by 30% by serving as Scrum Master for a 7-person Agile team, driving sprint planning, client communications, and rigorous PR reviews.
- Reduced deployment failures by 40% by architecting CI/CD pipelines via GitHub Actions and Vercel on Azure VM infrastructure, streamlining the team's release workflow.

AI Software Developer Intern

JuliGerm

Lesotho, South Africa (Remote)

Apr 2025 - Aug 2025

- Reduced system downtime by 30% by designing and maintaining AI-powered backend pipelines on Microsoft Azure to process agricultural and satellite data at scale.
- Cut farmers' manual tracking effort by 40% by building a custom search engine that unified crop health, soil quality, water usage, and weather data into a single decision dashboard.

AI, Software Developer

MedXAI

Edmonton, AB

July 2024 - August 2025

- Improved hospital resource planning by enabling patient load predictions 2 weeks earlier, achieving 90% forecast accuracy by implementing Python-based generative and classification models on COVID-19 data.
- Reduced researcher reporting time by 25% by building interactive dashboards and analyzing 100+ datasets to visualize real-time and forecasted trends, strengthening model trust and accuracy.

ML, Web Development Intern

UAIS (Undergraduate Artificial Intelligence Society)

Edmonton, AB

January 2024 - April 2024

- Delivered 85% accurate industrial risk analysis by deploying NLP pipelines, reducing manual review workload by 40% and improving prediction reliability by 20% and improving model runtime by 10%.
- Automated report generation with OpenAI API and applied PCA with feature scaling on large datasets, cutting client delivery time by 40%.

PROJECTS

Field Notes AI | React Native, Supabase, Azure, SonarQube, Jest

Team Project

April 2026

- Maintained zero critical security vulnerabilities across the codebase by enforcing strict quality gates using ESLint, CodeQL, and a comprehensive SonarQube integration for a cross-platform AI app serving social workers.
- Ensured highly reliable and secure delivery by architecting a scalable self-hosted Supabase backend and automating APK build pipelines, cutting manual release effort significantly.

BloodBridge: Blood Donor Platform | React, Next.js, TypeScript, Supabase, PostGIS

HackED 2026

January 2026

- Reduced donor match time by leveraging PostGIS proximity queries and Supabase Realtime push notifications to build a real-time platform connecting hospitals with compatible blood donors.
- Improved match precision for critical cases by designing a geo-aware engine incorporating blood-type compatibility filtering, urgency triage, and live donor availability tracking.

CampusNest: Student Community App | React Native, Expo, Supabase, Tailwind CSS

Personal Project | Github Repo: github.com/campus-nest/campusnest

April 2026

- Centralized campus community engagement by building a cross-platform mobile app in React Native and Expo, supporting secure user profiles, item listings, and real-time activity feeds.

- Ensured reliable application deployment by integrating Supabase for authentication and data syncing, and establishing automated CI/CD pipelines with Jest test coverage.

Mood Tracker Android App | Java, Android SDK, Firebase, Google Maps SDK

Team Project | Github Repo: github.com/cmput301-w25/project-kernelcrew

April 2025

- Enabled 100+ users to reflect on emotions more consistently by building an Android mood-tracking app with geotagged pins, custom timestamps, and offline sync capabilities.
- Increased daily log completion rates by 25% during testing by implementing authentication, social following, and push notifications in a 6-member Agile team.

BearBazaar: Campus Marketplace | React.js, Express.js, Prisma, SQLite, Node.js, REST API

HackED 2025 | Github Repo: github.com/adityadipakpatel/BearBazaar

February 2025

- Supported secure listings and real-time messaging for 200+ students by building a full-stack marketplace with a React.js frontend and an Express.js + Prisma backend.
- Reduced average API response time by 20% by optimizing SQL queries and REST endpoints, ensuring reliable real-time updates across the platform.

ConsoleTweeter: CLI Social Platform | Python, SQLite, MongoDB

Team Project | Github Repos: [SQLite](#) · [MongoDB](#)

December 2024

- Validated cross-backend correctness on 200+ sample tweets by engineering a CLI social platform with authentication, tweeting, search, and following across both SQL and NoSQL backends.
- Improved search precision and user navigation efficiency by 30% by implementing hashtag parsing and contextual user filters.

MatchHire: Resume Builder | Python, JavaScript, ChatGPT API, PdfLatex API

GHacks 2024

August 2024

- Delivered a fully functional resume auto-generation tool in 24 hours by leading a 4-member team to build a resume parser and LaTeX PDF generator via a web interface.
- Reduced manual formatting time by 80% by integrating LLM APIs with Python and PdfLatex to automate resume structuring and PDF rendering.

RareQuest: Disease Awareness Game | HTML, CSS, JavaScript

Code to Give (Morgan Stanley) 2024 | Github Repo: github.com/adityadipakpatel/RQMO12

May 2024

- Engaged 50+ players in rare disease awareness by creating a browser-based game featuring interactive diagnosis challenges built with HTML, CSS, and JavaScript.
- Improved player retention and cut load time by 20% by optimizing JavaScript game logic and database operations for smoother gameplay performance.

Sign Language Detector | OpenCV, TensorFlow, Python, Jupyter Notebooks

HackED 2024 | Github Repo: github.com/adityadipakpatel/Sign-Language-Detector

January 2024

- Achieved 90% gesture-to-text translation accuracy by building a real-time sign language detection app using TensorFlow and OpenCV.
- Enabled faster digital communication for Deaf and Mute individuals by reducing recognition latency to under 200ms per gesture through model optimization.

EXTRACURRICULAR PROJECTS

CryptoHash: Cryptographic Algorithm (SHA256) Implementation | Python, Cryptography, Cybersecurity

- Deepened applied cryptography knowledge by implementing the SHA256 algorithm from scratch in Python to generate standards-compliant 256-bit hash values.
- Achieved 100% cross-platform reliability by validating full compatibility with Unix/Linux SHA256 utilities through system-level integration tests.

WeatherSage: Weather Data Retriever | Python, OpenWeatherMap API, JSON

- Automated real-time weather retrieval across multiple locations by building a Python tool that fetches, parses, and visualizes data from the OpenWeatherMap API.
- Reduced manual data lookup time by 80% by transforming raw JSON API responses into structured, user-readable outputs.

SecureText: Text Encryption Utility | Python, Cryptography, Security

- Protected sensitive user data by developing a two-passcode encryption and decryption system in Python, ensuring end-to-end privacy.
- Maintained 100% data integrity across all transmission tests by applying cryptographic algorithms to encode text into numeric strings and restore originals securely.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: ReactJS, NodeJS, ExpressJS, Redux, Django, Flask, PyTorch, TensorFlow, scikit-learn

Databases & Cloud: PostgreSQL, MySQL, MongoDB, Microsoft Azure, Amazon Web Services, Google Cloud Platform

Developer Tools: Git/GitHub, Postman, Docker, Android Studio, Linux, Windows